

THE STANDARD ERROR OF A HISTOGRAM FROM A STATIONARY TIME SERIES

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1. Introduction

Histograms obtained from stationary time series data can be used to form an opinion on the probability density function. For *independent* observations the number of observations falling into a given interval has a binomial distribution, and a standard error can be found accordingly. Here we obtain, under the assumption of *stationarity* the covariance of frequencies in two intervals, which includes, as special case, the variance of a frequency in one interval, from which a standard error can be calculated.

In fact we solve here a problem which is slightly more general than would be necessary for finding the standard error of a frequency. This more general result will be applied in a subsequent paper to find the standard error of estimators of the intensity function of stationary point processes.

As will be seen the variance of a frequency depends on as many parameters as there are observations, so the number of parameters has to be reduced. This is achieved here by basing the estimators on a variance formula which would be correct if X_i and X_j were independent for $|i-j| > d$, $d=0,1,2,\dots$

2. Notation

Let X_i , $i \in I$, be a stationary stochastic process.

We shall use $F_r(u)$, $r=1,2,\dots, u \in \mathbb{R}$, to denote the distribution function of

$X_i + X_{i+1} + \dots + X_{i+r-1}$ and $F_{r(k)s}(u,v)$, $r,s=1,2,\dots, k=0, \pm 1, \pm 2,\dots, u,v \in \mathbb{R}$ to denote the joint distribution function of

$$X_i + X_{i+1} + \dots + X_{i+r-1} \text{ and } X_{i+k} + X_{i+k+1} + \dots + X_{i+k+s-1}.$$

We shall also write $f_r(u) = (F_r(u + \Delta u) - F_r(u))/\Delta u$, $\Delta u > 0$ for an approximation to the probability density function — if it exists — of the process $X_i + X_{i+1} + \dots + X_{i+r-1}$, $i \in I$, at the argument $u + \Delta u/2$.

Similarly, for an approximation of the bivariate density

$$f_{r(k)s}(u,v) = (F_{r(k)s}(u + \Delta u, v + \Delta v)$$

$$- F_{r(k)s}(u + \Delta u, v) - F_{r(k)s}(u, v + \Delta v) + F_{r(k)s}(u, v)) / \Delta u \Delta v.$$

For the empirical density function of $X_i + X_{i+1} + \dots + X_{i+r+1}$ (calculated from a realisation of size n), that is for $(1/\Delta u)$ times the relative frequency of observations in the interval $(u, u + \Delta u]$, we shall write ${}_n f_r(u)$.

$$\begin{aligned} \text{Setting } y_i^{(r)}(u) &= 1 \text{ if } u < X_i + \dots + X_{i+r-1} \leq u + \Delta u, \\ &= 0 \text{ otherwise,} \quad \text{let} \end{aligned}$$

$${}_n f_r(u) = \frac{1}{\Delta u(n-r+1)} \sum_{i=1}^{n-r+1} y_i^{(r)}(u) = \frac{1}{\Delta u(n-r+1)} w(r; u).$$

In the bivariate case we shall write

$$\begin{aligned} {}_n f_{r(k)s}(u,v) &= \frac{1}{\Delta u \Delta v n^*} \sum y_i^{(r)}(u) y_{i+k}^{(s)}(v) \\ &= (1/\Delta u \Delta v n^*) z(r, s; k; u, v), \text{ say.} \end{aligned}$$

The sum is taken over all possible subscripts i , for the given set of data. Thus $n^* = n^*(n, r, s, k)$, the number of terms in the sum, depends on n, r, s and k . For all further derivations it is preferable to use n^* instead of its explicit value in terms of n, r, s, k and to note that for $s \leq r$

$$\begin{aligned} n^* &= n - k - s + 1 & \text{for } r - s \leq k \\ &= n - r + 1 & 0 \leq k \leq r - s \\ &= n + k - r + 1 & k \leq 0 \end{aligned}$$

and for $r \leq s$

$$\begin{aligned} n^* &= n - k - s + 1 & \text{for } 0 \leq k \\ &= n - s + 1 & r - s \leq k \leq 0 \\ &= n + k - r + 1 & k \leq r - s \end{aligned}$$

We shall also use

$$K(n, r, s, d) = (n - r - s - d + 1)(n - r - s - d + 2) / (n - s + 1)(n - r + 1) - 1$$

and note that

$$E_n f_r(u) = f_r(u) \text{ and } E_n f_{r(k)s}(u, v) = f_{r(k)s}(u, v).$$

3. The covariance of two empirical densities

We calculate now $\text{Cov}(f_r(u), f_s(v))$ and assume that $s \leq r$. To determine the standard error of a histogram, i.e. the standard error of the empirical density of X_1 itself, it is only necessary to consider the case $r = s = 1$.

One needs $E_{n f_r(u) f_s(v)}$ and we express at first $n f_r(u) f_s(v)$ more explicitly:

$$\begin{aligned} \Delta u \Delta v (n-r+1)(n-s+1) n f_r(u) f_s(v) &= \sum_{\mu=1}^{n-r+1} \sum_{\nu=1}^{n-s+1} y_{\mu}^{(r)}(u) y_{\nu}^{(s)}(v) \\ &= \sum_{\nu=0}^{r-s} \sum_{\mu=1}^{n-r+1} y_{\mu}^{(r)}(u) y_{\nu+\mu}^{(s)}(v) + \sum_{\nu=r-s+1}^{n-s} \sum_{\mu=1}^{n-s+1-\nu} y_{\mu}^{(r)}(u) y_{\nu+\mu}^{(s)}(v) \\ &\quad + \sum_{\nu=1}^{n-r} \sum_{\mu=1}^{n-r+1-\nu} y_{\nu+\mu}^{(r)}(u) y_{\mu}^{(s)}(v) \\ &= \sum_{\nu=0}^{r-s} z(r, s; \nu; u, v) + \sum_{\nu=r-s+1}^{n-s} z(r, s; \nu; u, v) + \sum_{\nu=1}^{n-r} z(s, r; \nu; u, v) \\ &= \sum_{\nu=-(n-r)}^{n-s} z(r, s; \nu; u, v), \end{aligned}$$

since $z(s, r; \nu; u, v) = z(r, s; -\nu; u, v)$.

$$\text{Because } E z(r, s; \nu; u, v) / \Delta u \Delta v = E n^* f_r(\nu-r) s(u, v) = n^* f_r(\nu-r) s(u, v)$$

one has

$$\begin{aligned} (1) \quad \text{Cov}(n f_r(u), n f_s(v)) &= \frac{1}{(n-r+1)(n-s+1)} \sum_{\nu=-(n-r)}^{n-s} n^*(n, r, s, \nu) f_r(\nu) s(u, v) \\ &\quad - f_r(u) f_s(v) \quad (s \leq r). \end{aligned}$$

In particular:

$$(2) \quad \text{Var}(n f_r(u)) = f_r(u) / \Delta u (n-r+1) + (2 / ((n-r+1)^2)) \sum_{\nu=1}^{n-r} (n-r+1-\nu) f_r(\nu) f_r(u, u) - f_r^2(u)$$

4. Estimation of the covariance of empirical densities

One would obviously try to estimate the variance of $n f_r(u)$ by replacing, in the obtained formula $f_r(\nu) f_r(u, u)$ by $n f_{r(\nu)}(u, u)$ and $f_r(u)$ by $n f_r(u)$.

$$\text{Now } \sum_{\nu=-(n-r)}^{(n-r)} n^* n f_{r(\nu)}(u, u) = \frac{1}{(\Delta u)^2} \sum_{\nu=-(n-r)}^{(n-r)} z(r, r; \nu; u, u)$$

$$= {}_n f_r^2(u) (n-r+1)^2.$$

The estimator of the variance would thus be equal to zero. This is so because the number of parameters to be estimated is equal to the number of observations. The number of parameters can only be reduced by introducing new assumptions. For most stationary time series the covariance function tends to zero for large lags and one could assume that X_i and X_j are independent for $|i-j| > d$.

Under this assumption $f_r(\nu)_s(u, v) = f_r(u) f_s(v)$ for $\nu \geq d+r$ and for $\nu \leq -s-d$, and

$$\begin{aligned} \text{Cov}({}_n f_r(u), {}_n f_s(v)) &= \frac{1}{(n-r+1)(n-s+1)} \sum_{\nu=-s-d}^{-s-d} {}_n f_r(u) f_s(v) + \sum_{\nu=-s-d+1}^{d+r-1} {}_n f_r(\nu)_s(u, v) \\ &\quad + \sum_{\nu=d+r}^{n-s} {}_n f_r(u) f_s(v) - f_r(u) f_s(v) \\ &= \frac{1}{(n-r+1)(n-s+1)} \sum_{\nu=-s-d+1}^{d+r-1} {}_n f_r(\nu)_s(u, v) + K(n, r, s, d) f_r(u) f_s(v) \end{aligned}$$

An estimator of the covariance is then:

$$\begin{aligned} (3) \quad \hat{\text{Cov}}({}_n f_r(u), {}_n f_s(v)) &= \frac{1}{(n-r+1)(n-s+1) \Delta u \Delta v} \sum_{\nu=-s-d+1}^{d+r-1} z(r, s, \nu; u, v) \\ &\quad + K(n, r, s, d) {}_n f_r(u) {}_n f_s(v), \quad s \leq r \end{aligned}$$

and in particular an estimator of the variance:

$$(4) \quad \hat{\text{Var}}({}_n f_r(u)) = \frac{1}{(n-r+1)^2 (\Delta u)^2} (w(r; u) + 2 \sum_{\nu=1}^{d+r-1} z(r, r, \nu; u, u) + K(n, r, r, d) w^2(r; u))$$

Estimators of the variance and covariance of empirical densities of a histogram (of X_j itself) are obtained by setting $r=s=1$. To get the corresponding results for frequencies one has to multiply the results by $n^2 (\Delta u)^2$.

5. Experiments

The method was tried out in very many experiments on autoregressive, moving average and autoregressive-moving average processes with various parameters and sample sizes. It is impossible to list the results, or even a representative selection of them, without using too much space. We only summarise the impressions gained qualitatively, and illustrate them.

It is clear that using small values of d in (4) leads to biased estimators. The bias

depends on the correlation function of the process used, it decreases with increasing d . The value of d for which the bias becomes negligible is larger for processes with correlation functions which persist. The standard error of the estimators, however, increases very rapidly with increasing d and this militates against using values of d which would seem optimal with respect to bias. If mean squared error is used as criterion then the optimal value of d for small lag one correlations is zero or one, and for larger lag one correlations one or two. Large sample sizes also tend to push up the optimal value of d . (In our experiments the lag one correlations ranged between ,2 and ,8 and the sample sizes between 100 and 1 000.)

So it turns out that it is usually not possible to exploit the exact result (2) fully. The basic difficulty of all *general* time series methods is strongly felt here as well: the need to estimate a large number of parameters. Unless the sample size is really large, or unless one wants to specialise and assume families of processes with a few parameters only, one has to find and use rough approximations.

The roughest approximation of all, $d=0$, is equivalent to the classical standard error formula for independent observations. It should be applied if the absolute value of the lag one correlation is smaller than ,2. In all other cases taking $d=1$ will be adequate unless the absolute lag one correlations, or the sample size, or both, are very large. The table below may serve as a guide. It gives the estimated bias and the estimated mean square error of estimators of the standard deviation of frequencies (based on (4)) in the interval (0,1). The process used was autoregressive of order one with parameter ϕ , the sample size is denoted by n . The correlation function of this process is $\phi^{|r|}$, the variance was 1,04 for $\phi = ,2$ and 2,78 for $\phi = ,8$. The expected frequencies in (0,1) are ,34 n and ,23 n respectively. To obtain the estimated bias the standard deviation of frequencies was estimated directly from 200 independently obtained histograms. The difference between this empirical standard deviation and the average of 50 independent estimators based on (4) is the estimated bias given in the table. The estimated mean squared error is (estimated bias)² + (estimated variance), where the estimated variance was also calculated from 50 independent estimators based on (4).

d		0		1		2		3	
n	ϕ	BIAS	MSE	BIAS	MSE	BIAS	MSE	BIAS	MSE
100	,2	,71	,55	,57	,82	,74	1,34	,82	2,84
200	,2	1,18	1,42	1,03	1,26	,99	1,39	1,37	3,47
400	,2	1,19	1,71	,90	2,20	,90	4,03	1,07	10,02
1000	,2	1,70	3,78	1,33	8,40	1,23	13,32	1,40	29,92
100	,8	1,75	3,23	1,10	1,66	,86	2,38	,64	3,35
200	,8	2,20	5,01	1,42	2,43	,90	1,48	,75	1,65
400	,8	2,82	8,96	1,52	4,79	1,18	5,02	,52	8,04
1000	,8	5,22	29,63	3,23	15,89	2,08	14,81	1,13	32,96